

ESSIC Summer Internship Program: End of Summer Report

Project Title: Validation of NOAA Satellite Vegetation Phenology

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Section 0: Background

Vegetation phenology, the timing of seasonal ecological cycles such as spring green up and fall senescence, is a critical variable for the National Centers for Environmental Prediction (NCEP). Accurate phenology data is required for numerical weather prediction and climate models to correctly simulate the exchange of water, and carbon between the Earth's surface and the atmosphere. To monitor this globally, NOAA utilizes the satellite derived Green Vegetation Fraction (GVF) product captured by the Visible Infrared Imaging Radiometer Suite (VIIRS) instrument.

However, with early investigations, significant inconsistencies seem to exist between modeled vegetation fractions, satellite observations, and ground truth data. Climate models often simulate slow, gradual seasonal transitions, whereas actual ground level ecological shifts are much more abrupt. Previous NOAA validation efforts identified systematic biases in the satellite data itself: GVF often exhibits phenological "compression" (an artificially shortened peak season) in mixed forests, and "lag" (a delayed detection of spring) in shrublands when compared to ground based PhenoCam sensors.

Validating these discrepancies is inherently difficult due to severe scale and sensor mismatches. A PhenoCam measures visible greenness (Green Chromatic Coordinate, or GCC) across a localized ~200m field of view, while VIIRS GVF measures structural canopy density across a large 1km spatial pixel. If a 1km satellite pixel contains a patchy mix of trees, roads, and lakes, it becomes impossible to tell if a timing "lag" is caused by a flaw in the satellite algorithm or simply a heterogeneous landscape. This project was initiated to build an automated, mathematically rigorous system to solve this spatial mismatch and confidently validate NOAA's satellite phenology products.

Section 1: Objectives

- Validate satellite phenology by comparing NOAA's VIIRS Green Vegetation Fraction (GVF) satellite data against ground based PhenoCam Green Chromatic Coordinate (GCC) data to quantify discrepancies (e.g., "lag" in shrublands or "compression" in mixed forests).
- Enforce spatial uniformity with the picked site to validate by establishing a "Golden

Standard" control group of highly uniform sites, ensuring that timing discrepancies are caused by algorithm lag rather than spatial mismatches

Section 2: Accomplishments

To investigate inconsistencies between satellite derived vegetation data from NOAA and weather models, I built an automated Python validation system. Previous research identified that satellite GVF often exhibits phenological "compression" or "lag" when compared to ground truth PhenoCam data. My primary task was to scale up this validation, evaluating all 1,281 PhenoCam site-years data across 2023 and 2024 to ensure we were comparing the sensors objectively.

1. Spatial Uniformity Filtering via Google Earth Engine

Land that is not uniform can drastically impact the testing of phenophase transition dates. It causes a major point of confusion: is the error actually derived from NOAA's VIIRS GVF algorithm, or is it just caused by the spatial mismatch between the satellite data and the ground based PhenoCam data?

A VIIRS satellite pixel covers a 1km x 1km area, sometimes averaging together forests, water, roads, and other surface types, while a PhenoCam observes a highly specific ~200m localized field of view. To solve this and ensure a valid comparison, the first step was to perform spatial uniformity filtering via Google Earth Engine. I used the GEE API to automatically evaluate spatial uniformity across thousands of PhenoCam observations and the GBOV (Ground-Based Observations for Validation) network, explicitly filtering out any land that had high fractions of water or urban areas.

For each candidate site, the computing pipeline centered a ± 30 day window at peak summer greenness (the plateau between the NDVI Maturity, MOS and Decline, DOS dates). It pulled cloud free Sentinel-2 composites ($< 5\%$ cloud cover) and calculated the spatial Coefficient of Variation ($\text{stdDev}/\text{mean}$) of the vegetation index across the 4km satellite footprint. Sites were evaluated against strict thresholds: spatial variance ($\text{CV} > 0.2$) and surface type contamination (Water $> 10\%$, Urban $> 10\%$) triggered automatic rejection. This successfully eliminated landscape patchiness, filtering the 1,281 site-years down to 320 highly uniform "Golden" site-years.

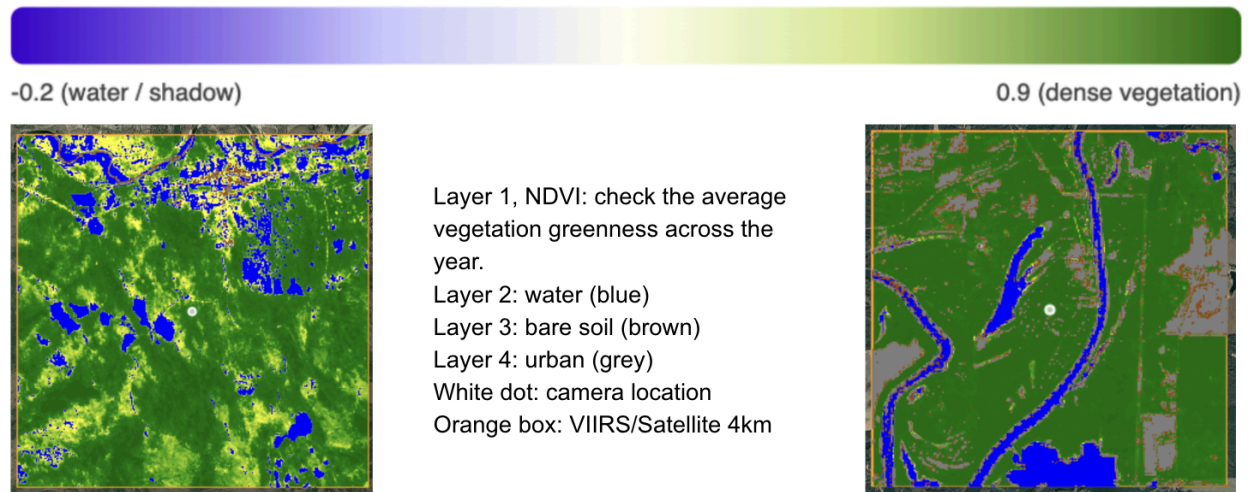


Figure 1: Google Earth Engine visualization comparing a spatially uniform site (left: *alercecosteroforest_EN_1001*) that passed the pipeline filters, versus a heterogeneous site (right: *DELA_DB_1000*) rejected due to high CV, water, and urban fractions within 4km.

Upon reviewing the 320 spatially uniform sites, I identified significant geographic clustering. I filter out duplicate camera locations and highly proximal sites. This yielded the final, geographically diverse subset of 40 'Golden Sites' distributed across the United States.

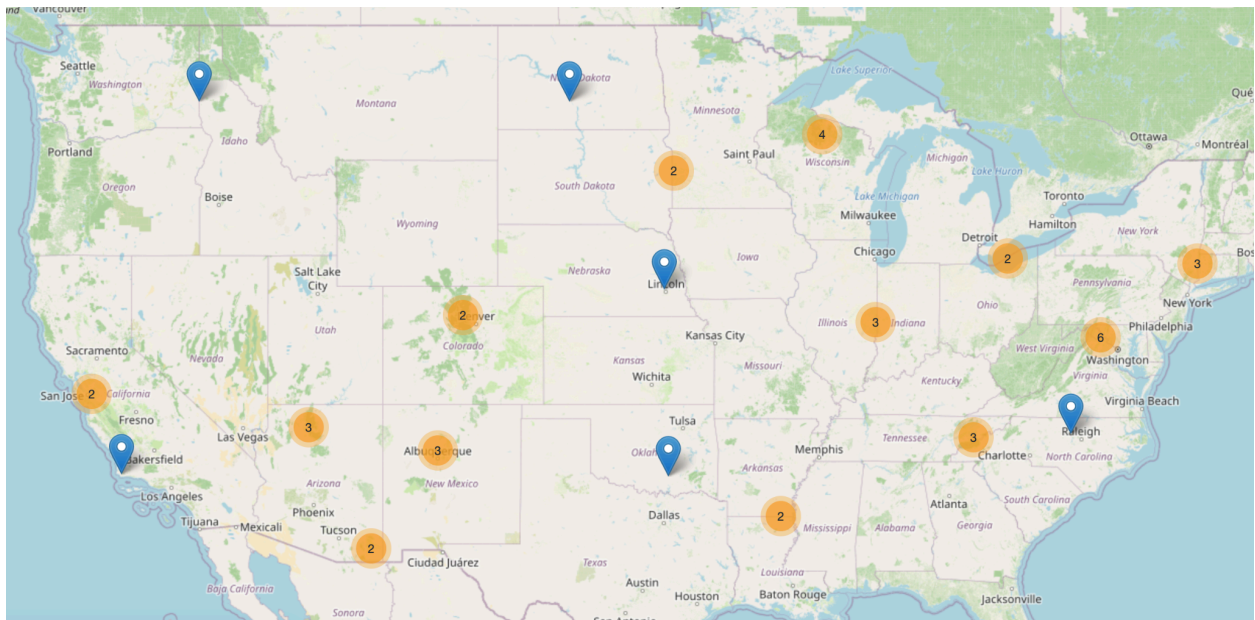


Figure 2: html map visualizations of the 40 sites distributions

2. Phenophase Extraction using Piecewise Logistic Regression

To extract the transition dates, the Curvature Change Rate (CCRmax) method was used (Zhang et al., 2003). The pipeline fits a piecewise double logistic function to both the GVF and GCC time series: one curve for the rapid spring green up, and a second for the slower fall senescence. By calculating the extrema of the analytical rate of change of curvature, the algorithm can precisely pinpoint the Start (SOS), Maturity (MOS), Decline (DOS), and End of Season (EOS).

3. Time Series Comparison via Dynamic Time Warping (DTW)

Standard statistical correlation will fail to accurately compare seasonal curves because it cannot account for biological lags. For example, if a satellite and a camera record the exact same shape, but the satellite is shifted two weeks late, basic cross-correlation will falsely score them as a poor match.

To solve this, I implemented Dynamic Time Warping (DTW). DTW is an algorithm that "warps" the time axis to align peaks and valleys, measuring the true shape similarity between two shifted curves. This method follows the industry standard used by the creators of PhenoCam to successfully align vegetation data from multiple different instruments (Richardson et al., 2016).

To rank the final 40 uniform sites, I calculated a "Divergence Score" by blending two metrics:

1. Phenophase Gap: The absolute average difference in days between the satellite and camera transition dates.
2. DTW per Step: Both curves are interpolated to a daily axis and min max normalized to [0, 1]. Dynamic Time Warping finds the optimal non-linear alignment between the two curves to score their overall shape similarity.

Thus form 3 metrics that can help us read the similarity between NDVI (satellite data), GCC (groundbase), and GVF (NOAA's VIIRS) time series.

- NDVI vs GCC divergence
- NDVI vs GVF divergence
- GCC vs GVF divergence

The results confirmed that even in perfectly uniform environments, genuine sensor divergence occurs. The pipeline also successfully identified "Golden Sites" with extremely low NDVI-GCC divergence where GVF seem to differ. See sites below:

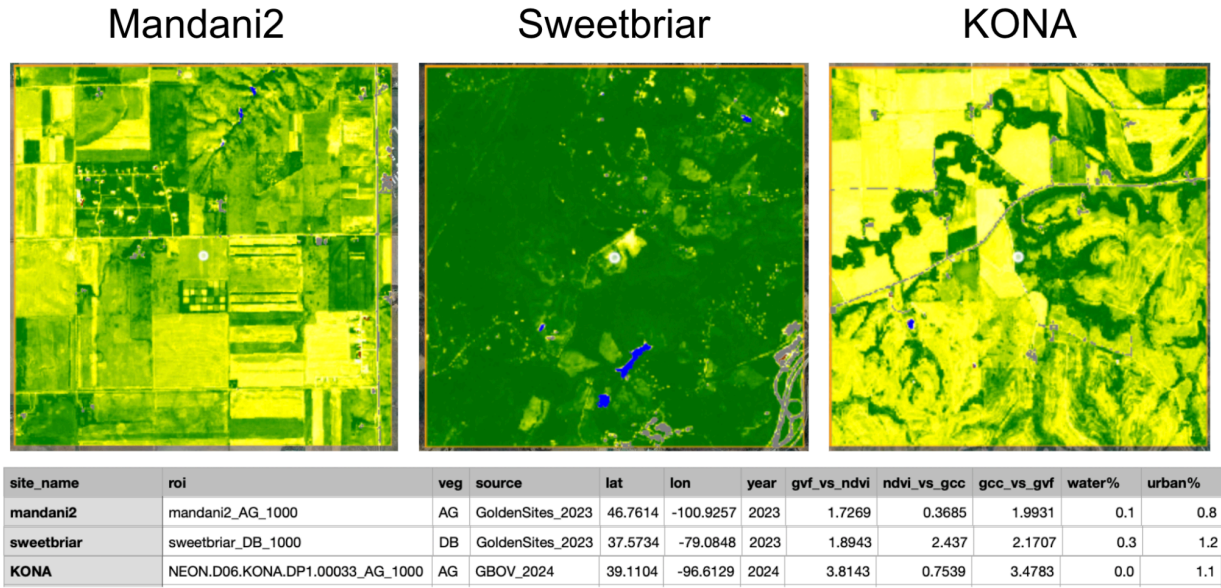


Figure 3: Google Earth Engine visualization with table of their respective 3 divergence scores, water% and urban%

By enforcing spatial uniformity, this project can mathematically confirm that discrepancies between satellite GVF and ground based PhenoCams are caused by genuine differences in either sensor physics, camera positioning, or the VIIRS algorithm.

4. Data Anomalies

After rigorous statistical testing on our "Golden Sites", the data reveal several vital insights:

1. "Stretching" within green up phase: To mathematically test whether satellite curves were shortened, I calculated phase duration ratio comparing the length in days of the satellite's green up transition against the ground camera's transition ($[GVF_MOS - GVF_SOS] / [GCC_MOS - GCC_SOS]$). In this ratio, a value < 1.0 indicates compression, while a value > 1.0 indicates stretching. Upon analysis, Phase duration ratios (GVF green up length / GCC green up length) consistently exceeded 1.0 for DB forests. Instead of observing "compression" from the original hypothesis, the satellite GVF curve is "stretched" and smoothed out, failing to capture the abrupt data documented by ground cameras.

GBOV + GoldenSites — Compression 2023 (blue) vs 2024 (green)

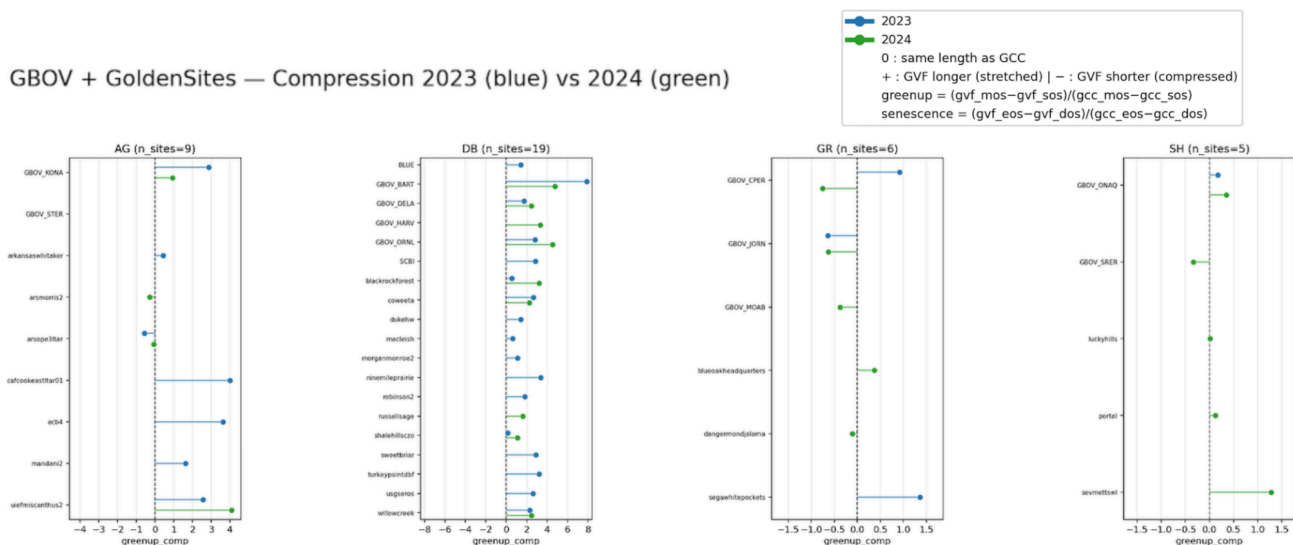
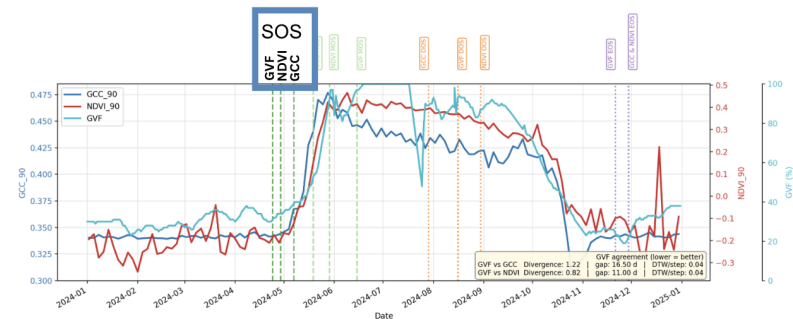
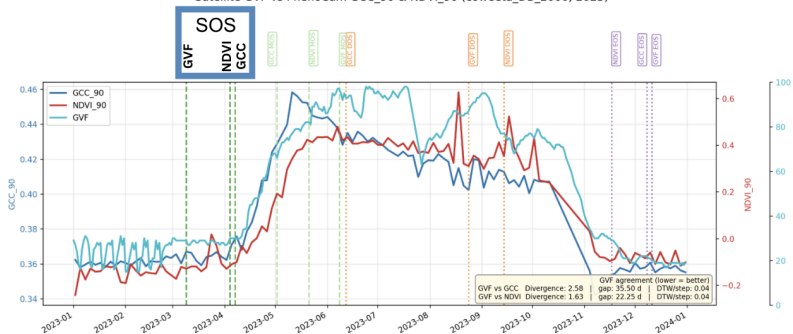


Figure 4: lollipop plot of 2023 and 2024's green up phase's compression analysis on different land cover

Satellite GVF vs PhenoCam GCC_90 & NDVI_90 (NEON.D01.HARV.DP1.00033_DB_1000, 2024)



Satellite GVF vs PhenoCam GCC_90 & NDVI_90 (coweeta_DB_2000, 2023)



GBOV + GoldenSites — Phase lag 2023 (blue) vs 2024 (green)

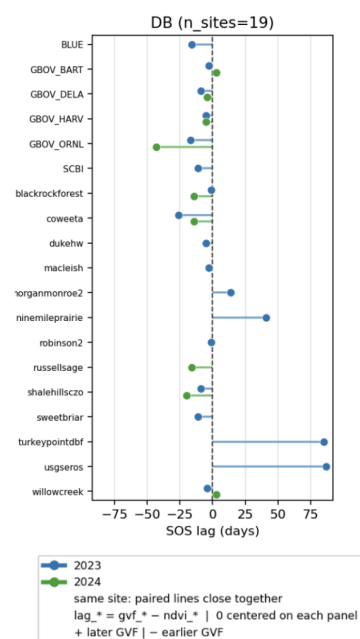


Figure 5: The two graphs on the left show the time series of GVF, GCC, and NDVI of two sites that have an early SOS date. The lollipop graph on the right shows all the sites-year of SOS date lag of the deciduous broadleaf landcovers.

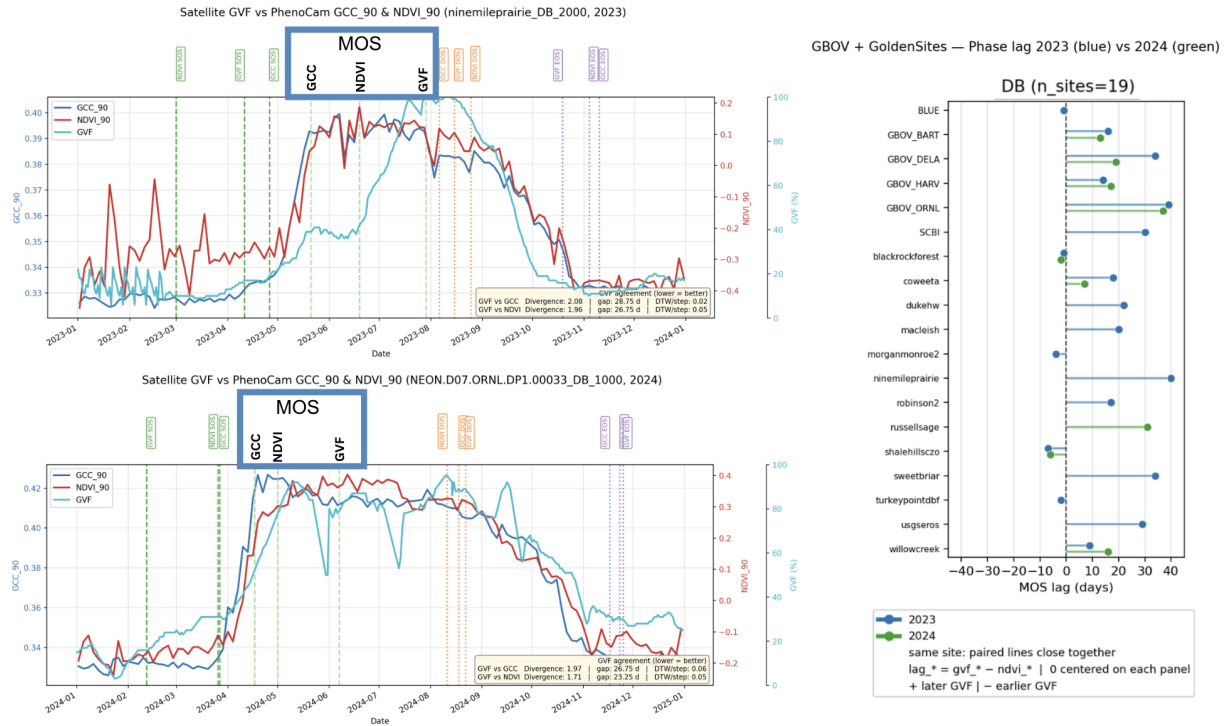


Figure 6: The two graphs on the left show the time series of GVF, GCC, and NDVI of two sites that have late MOS dates. The lollipop graph on the right shows all the sites-year of MOS date lag of the deciduous broadleaf landcover.

2. The early SOS date phenomenon: At uniform DB sites, the satellite (GVF) frequently detected the Start of Season (SOS) earlier than the camera (see right side of figure 5, where there are multiple small amounts of negative lag). I hypothesize this is a geometric sensor limitation: satellites looking straight down (nadir) detect the early green-up of the forest floor's understory, while ground cameras do not register SOS until the upper tree canopy blooms weeks later. In contrast, the satellite frequently detected the Middle of Season (MOS) later than the camera for DB sites. (see right side of figure 6 where there are multiple positive lag). Interestingly together, this perfectly explains the "Stretching" phenomenon of part 1's finding. At the start of the season GVF detects earlier and then for the middle of the season, GVF detects later.

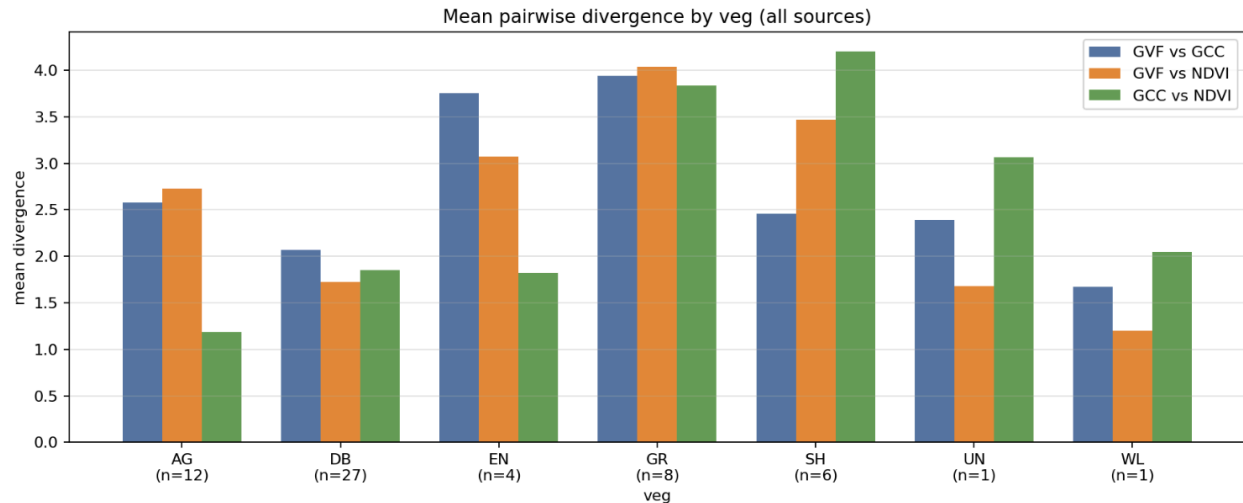


Figure 7: Bar Graph of mean of all site's 3 divergence across all type of landcover

- Statistically Validating Sparse Biome Lag: A one sided Welch t-test ($p = 0.007$) and a large Cohen's d effect size (1.21) proved that sparse landscape, GR site (Shrublands/Grasslands) suffer a mathematically significant detection delay, lagging an average of 22 days beyond the DB site baseline. Overall divergence is also higher for GR sites relative to DB sites, see figure 7.

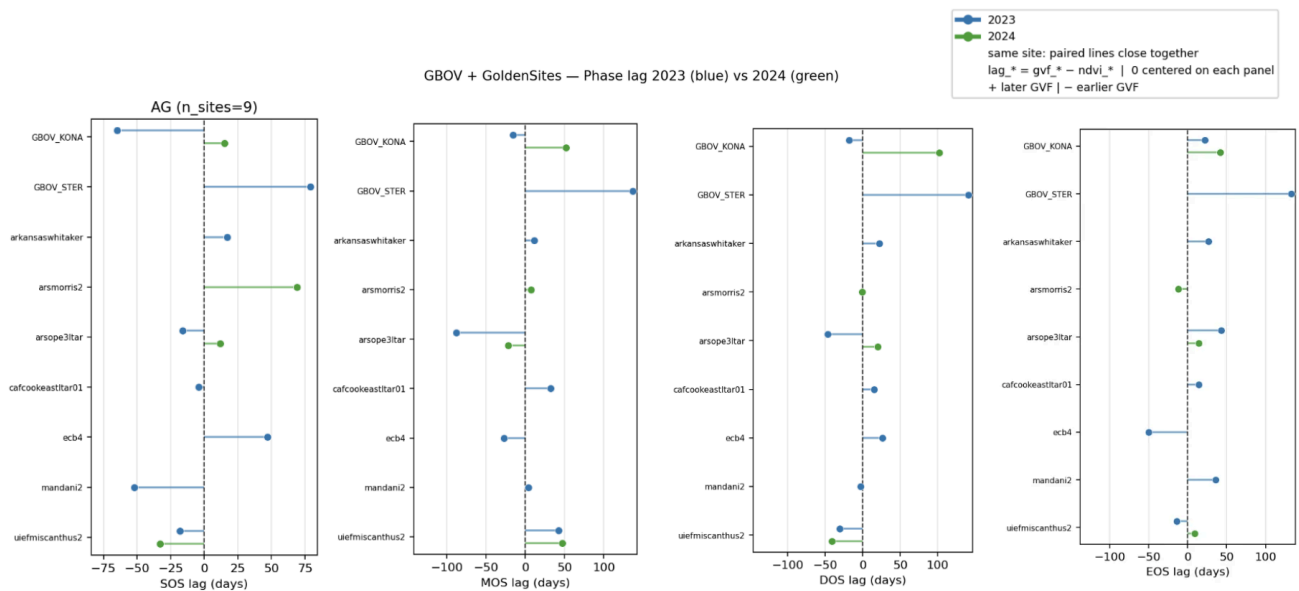


Figure 8: lollipop graph for AG landcover of four phenophase of lag

- Agricultural Land Cover Instability: Human managed lands demonstrated extreme phenological unpredictability (oscillating from -50 to +50 days of lag), confirming that sudden planting schedules and harvesting events violate the natural curve assumed by piecewise logistic models.

Section 3: Conclusions

The calculations proved that the satellite data actually stretches out the seasonal transitions rather than compressing them. This happens because of a two part problem. First, there is a possible geometric camera limitation where the satellite looks straight down and catches the early blooming, giving us a premature Start of Season compared to a possible unknown angle of the groundbase data, which might not see the full foliage. Second, the mathematical filters used in the satellite pipeline smooth out the data so much that it pushes the peak maturity date later into the summer.

The analysis revealed that satellite detection accuracy is highly biome dependent. Statistical testing confirmed that sparse landscapes (shrublands and grasslands) suffer a mathematically significant detection delay, lagging roughly three weeks behind deciduous forest baselines. Furthermore, human managed agricultural lands demonstrated unpredictability (with lags oscillating wildly from -50 to +50 days). Agricultural sites are inherently unreliable and not useful for this type of satellite validation analysis and will not be investigated further for the near future.

Section 4: Future Work

To understand the root causes of the stretching and lag I observed, my next steps will involve analyzing more data tailored specifically to Evergreen Needleleaf, Grassland, and Shrubland biomes. My initial data collection focused heavily on finding geographically spread out and independent sites across the United States. While this prevented spatial overlap, it also meant we did not have enough sample sites per biome to confirm clear patterns for those specific sparse landscapes. Gathering a larger pool of data for these specific environments will help clarify these vague trends.

Additionally, I will be investigating the reasons for “stretching” in the deciduous forest. I will be testing the camera angle vs satellite detections differences hypothesis by looking at the actual images produced by phenocam whenever the stretching phenomenon occurs. I will also investigate the official NOAA NVPS GVF operational pipeline. By evaluating specific stages of the algorithm flowchart, I plan to see if any of these mathematical processing steps actually cause the anomalies we found in the final satellite product.

References:

- Zhang, X. (2002, September 9). *Monitoring vegetation phenology using Modis*. elsevier. https://engr-didan02s.ece.arizona.edu/documents/dataviewer/Monitoring_Phenology_Zhang.pdf
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